

code	effect	required args	default color	color rotation	fade mode	sync	repeats	combinations	X used in shows	examples
48 00	prior color -> pulse twice, fade out prior effect ->pulsing	color/effect, prior 58 ??		depends upon prior	yes	yes	yes		0	95 24 64 58 12 48 00 ED
48 01	fade off	color/effect, prior 58 ??		no	yes	yes	no	lft side args: 58 (14,28,50)	45	94 F7 58 50 48 01 96
48 02	off, flash once, off	color		no	no	yes	no	58 ?? Args work	0	92 63 48 02 F6 - basic effect 94 63 48 02 58 64 84 - slow to begin, slow to fade
48 03	as one ear fades up, the other fades down	color, 58 F0	lft cyan, rt blue	no	yes	opposite	yes	lft side 58 50, rt side D0 42 (04,05,07,0A,0E)	217	95 24 67 58 F0 48 03 87 - alt red up, down 9B 20 26 A0 65 69 58 F0 48 03 D0 42 0A DF - magenta and blue fade up & down and switch ears
48 04	no arg, lft solid magenta, with two color args and 58 F0, like 48 03		lft arbitrary color? Sometimes magenta, blue, red,	no	arg dep	no	yes	58 F0 48 82	458	92 24 48 04 60 - left ear on solid magenta 94 24 58 F0 48 04 - left ear fades magenta up and down 96 20 64 6C 58 F0 48 04 50 - lft ear fades down as rt ear fades up then pattern reverses 92 24 48 82 31 and then the command
48 08	quick rotation through 4 colors			yes	no	yes	yes	either part of 26 29 03 1D 13 12 48 08 72 or with rt side of 48 8E or 72 ??	108	91 48 08 AC - cycles through white, blue, magenta, red 94 24 48 08 48 8E 2D - adds fade up, down to the color rotation
48 0D	mostly used for strobe-like effect	color, 58 1-6	no	no	no	yes	yes	lft side 58 (01,02,06)	126	95 24 67 58 01 48 0D - strobe like flashing white
48 0E	rapid alternating ear effect	color, 58 02	no	no	no	opposite	yes	lft side 58 02, often a right side arg for 48 08	55	98 20 24 61 0E 93 58 02 48 0E 6E - rapid alternating lft orange, rt blue
48 0F	slow, sync pulsing with args.	color, 59 0? ?? Left side	no	no	yes	yes	yes	lft side 59 0? ??, rt side D0 34 EE, D0 3A 32, 48 04, 48 82	208	96 24 61 59 02 07 48 0F 0D - slow, sync, blue, fade up and down both ears
48 10	no arg, two quick lft ear magenta flashes that repeats		magenta	no	no	no	yes	rt side D0 42 F3	7	97 F5 24 0D 0E 08 6D 48 10 1B - two quick magenta flashes, pause, repeat, lt and rt slightly out of sync 99 25 0D 66 48 10 D0 42 F3 FF 24 BB - rt
48 11	no args: two left yellow flashes, one rt yellow flash		yellow	no	no	no	yes	rt side D0 3D 01 or color code(S)	372	91 48 11 AD - basic pattern 95 F3 26 48 11 67 6F E7 - basic pattern, white color 99 F2 24 48 11 67 6F D1 41 09 07 F4 -

48 12	sync, quick flash, color rotation		red,yellow,blue,cyan	yes	no	yes	yes	no data	0	91 48 12 4F - basic pattern
48 13	quick white flashes, one side fades up while the other fades down, then		white	no	yes	opposite	yes	right side D0 3D (01,02)	142	98 FB 24 48 13 D0 3D 02 67 6F 6D - rapid white flashing both ears, fades up on one side, other fades down then reverses.
48 14	async color rotation of quick fade up, fade down		blue, green, orange,	yes	yes	no	yes	no data	0	91 48 14 92
48 15	sync, color rotation, fade up		red, green, yellow,	yes	yes	yes	yes	insufficient data	2	91 48 15 CC - basic pattern
48 16	sync, color rotation, fade up, fade down, color changes during fade		red,blue, magenta, off	yes	yes	yes	yes	no data	0	91 48 16 2E - basic pattern
48 17	sync, color rotation,		magenta, orange,	yes	no	yes	yes	no data	1	91 48 17 70 - basic pattern
48 18	async, color rotation, short, solid color in right ear, it switches to next color and it's previous color switches to left ear		orange,rose,light blue	yes	no	rotation	yes	no data	0	91 48 18 31 - basic pattern
*48 1C	?, sync, solid color - possibility this does nothing and is simply data for another argument	color, effect	none	no	no	yes	yes	48 83,48 84, left side 5B ?? ?? ??, rt side D0 39 06	77	92 67 48 1C - solid white. 98 24 67 58 02 48 84 F5 48 1C 0E - two quick white flashes then off
*48 1D	?, sync, color - possibility this does nothing and is simply data for another argument	color	none	no	no	yes	yes	left side D0 32 64, color, 5B A9 EE 2A 34	16	93 24 66 48 1D - solid yellow 9F FC 24 D0 32 64 69 5B A9 EE 2A 34 48 1D D0 3D 04 CB - lft ear flashes weak blue
48 1E	display off		none	no	no	yes	no	only found as simple 5 byte code. Only variation is col 2 and checksum	63	92 F6 48 1E D5 - used exclusively like this in show.
48 80	rapid rotation through color palette for one round and then slow fade up/down while colors rotate within the fades. Colors may be from extended color palette.		red, yellow, green, blue	yes	yes	no	yes	no data	0	92 24 48 80 8D - rapid 'flutter' one time through color palette and then fades up and down while colors rotate during the fades
48 81	rapid rotation through color palette and then out		green -> blue	yes	no	yes	no	no data	0	92 24 48 81 D3 - rapid rotation through colors one time. Then out.

48 82	rapid rotation through color palette and then out		green, cyan, blue, magenta, red, orange, yellow	yes	no	yes	yes	rt side args D0 ?? ??. not responsive to 58 ?? left hand side argument for 85, 84, 86, 0D, 0F	369	92 24 48 82 - basic pattern 99 F4 24 48 82 D0 0E FF D0 46 A0 5E - solid white. There are easier ways to get this effect, 91 24 67 9F. So...? 9A 9B 90 26 48 82 D0 0E 80 D0 42 1E B1 - slows the basic pattern down blue, magenta, red, orange, yellow, green, cyan 9A F1 0D 48 82 D0 0E FF 58 02 48 84 22 - strobe white
48 83	Synchronous smooth fade through the basic color palette (0x61->66)		red, yellow, green, cyan, blue	yes	yes	yes	yes	rt side D0 ?? ??, D0 0E 80,	585	91 48 83 00 - sync, smooth fade through basic color palette, minus 0x60 and 0x67 9B F4 24 46 81 EC 83 48 83 D0 0E A0 0D 75 - slow fade up, fade down (not off) through shades of blue. 9B F6 24 48 83 D0 0E 80 D0 42 03 0D A0 0E - more rapid cycling through the colors 9C 95 26 48 83 D0 0E 81 D0 42 14 D0 05
48 84	sync, repeating, short flashes, pause, roughly each second	color	none	no	no	yes	yes	left side 58 ??,rt side 5B A? F0 E6 33	326	92 24 48 84 EC - nothing 95 20 67 58 01 48 84 3C - rapid, sync, white flashing 93 24 64 48 84 BC - sync, red flashes, approx 1/sec
48 85	fade out	color, effect	none	no	yes	yes	no	lft side, 48 84, 48 86, D0 0E??, rt side 58 ??	2014	92 24 48 85 B2 - no effect, no default color. 93 24 64 48 85 E2 - no effect, 48 ?? code requires separation from leftside args by 0xF1-0xFF code. 94 24 64 F0 48 85 C7 - solid red, no fade. F0 not valid argument here 94 24 64 F2 48 85 88 - red then quick fade out. Intended effect. 94 48 85 61 58 32 EF - color follows fade codes. long blue fade out. 96 24 64 F2 48 85 58 32 DC - red, then very slow fade out. Addition of 58 argument 95 24 64 F2 48 85 61 48 - red, then blue fade out. Doesn't remain solid blue. 94 24 64 F2 48 85 88 followed immediately by 90 61 F8 - fails to give blue color. 94 24 64 F2 48 85 88 followed immediately by 91 24 61 42 - does result in solid blue.

48 86	fade up	color	no	no	yes	yes	no	rt side 58 ??, F? 48 85 rt side	593	92 24 48 86 50 - No effect, no default color. 93 24 64 48 86 00 - fade up red 93 24 48 86 64 12 - fade up red. color side doesn't matter 94 24 64 F0 48 86 25 - solid red. no fade up. F0 block 94 24 64 F1 48 86 8E - flash red and then fade up red. Different response to F1-F arg than 85. 94 61 58 32 48 86 58 - slow blue fade up 94 48 86 61 58 32 67- same above. arguments may follow the 48 86 command. 95 48 86 61 58 32 67 3D - slow white fade up. last color definition is the one used
48 87	async rotation, ears dif colors, yellow, green, cyan, blue, magenta, red, orange. repeats, rt ear leads rotation		yellow, green, cyan, blue, magenta,	yes	no	no	yes	left side 09 00 11 15 81 95 02, rt side D1 0E FF FF 72, 48 90		92 24 48 87 0E - basic pattern 97 F3 48 87 59 06 05 48 90 5D - rapid, async color rotation. 59 ?? ?? Timer for async effect. Second arg most influential. 9E FC 09 00 11 15 81 95 02 48 87 D1 0E
48 88	slow fades through red, yellow, green, blue, magenta. async but not by much, impression of rt to left fade, repeats		red, yellow, green, blue, magenta	yes	yes	no?	yes	lft side 20 76 72 ??, rt sid D0 0E FE D0 3D 20	56	92 24 48 88 4F - basic pattern 9B FA 24 48 88 D0 0E FE D0 3D 20 76 72 C3 - display off. Subsequent 0x24 needed to resume display 9D F6 24 29 11 1D 48 88 D0 0E FE D0 3D
48 89	sync blink. Roughly 1/sec	color	none	no	no	yes	yes	left side 5B 02 88 66 00, rt side 64 FD 24	13	93 24 67 48 89 A5 - basic pattern 9B F2 24 5B 02 88 66 00 48 89 64 FD 24 32 - two quick red flashes then off
48 8A	pause before starts, fade through colors, slightly async?		orange,yellow,green,cyan,blue	yes	yes	close?	yes	no data	0	92 24 48 8A F3 - basic pattern
48 8B	double flash, pause repeat	color	none	no	no	depends upon arguments	yes	lt side 5A ?? 08 55, rt side 58 01 48 0D	37	94 FD 24 67 48 8B C7 - sync, two white flashes, pause, repeat 95 FD 24 67 6F 48 8B D7 - alternating white flashes, two rounds, pause, repeat 99 FD 24 67 6F 5A 05 08 55 48 8B - rapid,
48 8C	rapid, constant flashing	color	none	no	no	yes	yes	no data	0	93 24 67 48 8C 9A - rapid, repeated
48 8D	rapid color rotation		magenta, blue, red, yellow, green. so quick hard to	yes	no	yes	yes	no data	0	92 24 48 8D 70 - basic pattern

48 8E	quick color rotation, combined with fade up and down with pause		fade up red,orange,yellow,cyan,green, blue,	yes	yes	yes	yes	no data	0	92 24 48 8E 92 - basic pattern
48 8F	display off			no	no	yes	no	no data	0	92 24 48 8F CC - display off. Blocking effect on some subsequent commands.
48 90	rarely used and requires substantial left side args. If it is a pattern, serves to speed up the argument's display				?	?	?	lft side 48 87 59 06 05	4	97 F2 48 87 59 06 05 48 90 1E - rapid, async color rotation
48 EC	not a stored effect code. Argument for D0 ?? ??								38	
48 EE	not a stored effect code. Argument for D0 ?? ??								22	
									5954	total codes covered
									11908	doubled to cover two bytes
									16.91	% code base described by this series